Intro-to-PDC Syllabus

(AI-centric version)

Spring 2026

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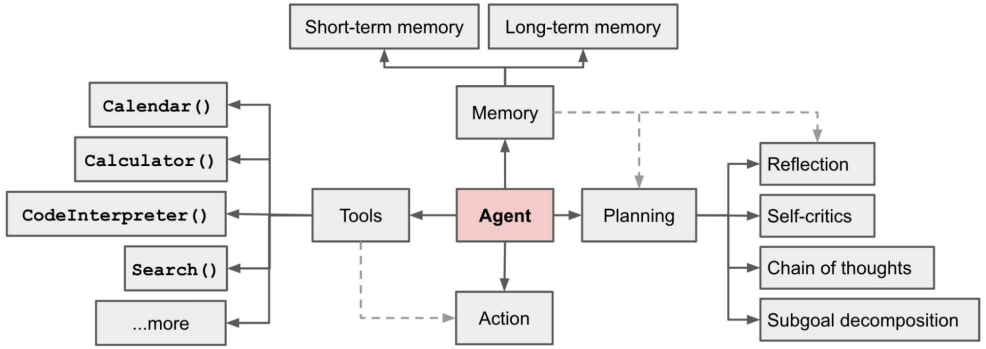
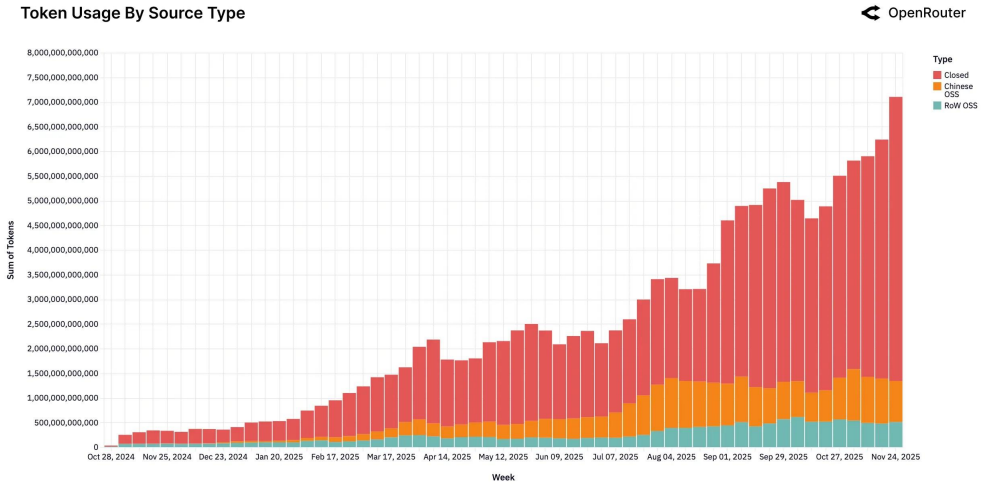
Token-Hungry Generative AI Applications

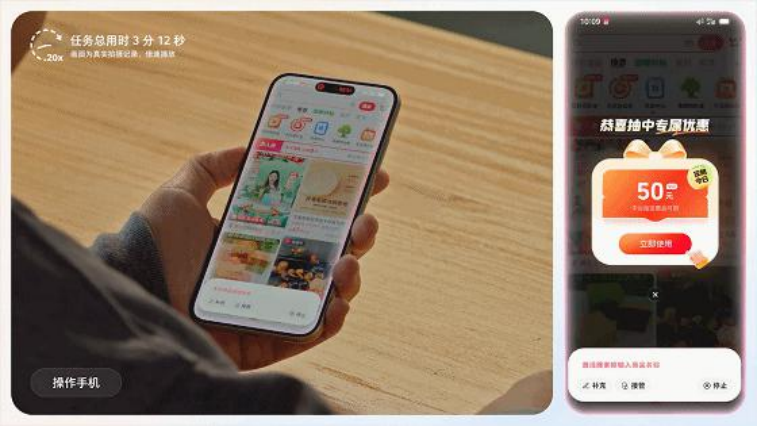
- LLM-powered autonomous agents



- Token consumption trends



- AI mobile phone and AI PC



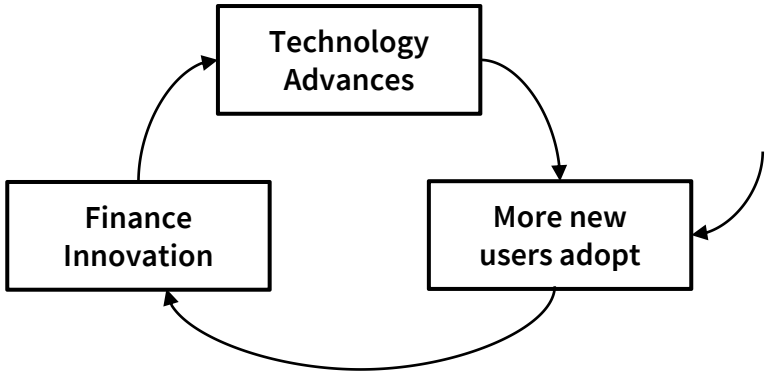
- Video generation

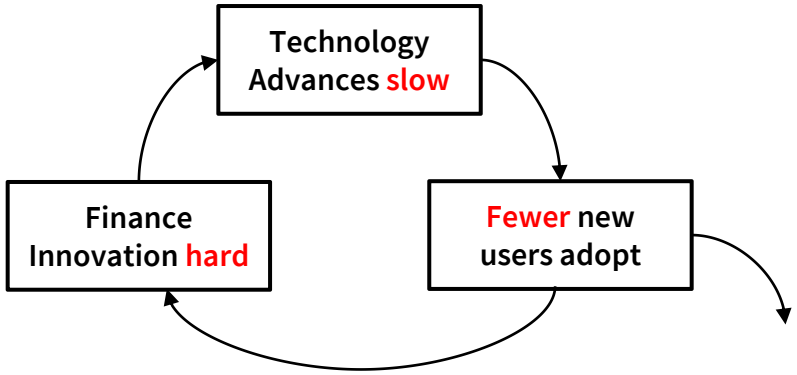
image source: lilianweng.github.io, openrouter.ai, 36kr.com (with doubao phone), github.com/baaivision/URSA

What is “General-Purpose” Computing?

- in terms of #programs → CPU is (still) general-purpose
- in terms of TOP/s → GPU/NPU are general-purpose

- Specialization fragments computing into “fast lane” and “slow lane”
  - “fast lane” applications that get powerful customized chips
  - “slow lane” applications that get stuck using general-purpose chips whose progress is fadin





B. N. C. Thompson and S. Spanuth, “The Decline of Computers as a General Purpose Technology,” MIT Initiat. Digit. Econ. Res. Br., vol. 1, pp. 1–6, 2019.

What were covered in the previous version?

- Questions to address
  - What is parallel and distributed computing?
  - How to do parallel & distributed programming, and do it well?
- Emphasis of this course
  - Basic concepts
  - Programming tools
  - Parallel algorithms
    - ("distributed" as in distributed-memory parallel system)
    - (NOT the general distributed system)
- Basic concepts
  - Hardware/software infrastructures
  - Design patterns
  - Performance models
- Programming tools
  - OpenMP; MPI; CUDA/HIP; NPU
- Algorithms
  - Matrix computation
  - Continuous optimization
  - Graph algorithms
  - Discrete search
- Emerging topics

What will be covered in the new version?

- Module 1: Fundamentals of AI System and Heterogeneous Computing
  - parallelism inside an GPU or NPU (the “device”), and between “host” and “device”
- Module 2: High-Performance AI Operators and Operator Engineering
  - parallelism within AI operators, as well as their typical optimizations
- Module 3: Principles and Optimization of AI Compilers
  - automatic or semi-automatic parallelization via compilation
- Module 4: Distributed Parallel Training & Inference and Communication
  - parallelism among compute cards and compute nodes
- Module 5: Performance Evaluation and AI-assisted Kernel Generation
  - next-generation of “automatic” parallelization

Grading & Class Schedules

- Grading (tentative)
  - Homework: 30%
  - Mini-project: 20%
  - Final exam: 50%
- Class meetings
  - Wed. 8am-10am (weekly), 三教501
  - Fri. 10am-12pm (bi-weekly), 三教501
- Office hours, by appointment
  - email: gluo@pku.edu.cn
  - wechat: luoguojie